Exploring the Barriers to Undergraduate Student Engagement with Research Papers: A Machine Learning and Statistical Analysis Approach

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***Abstract*— University students often have significant obstacles to participating in research publications, motivated by characteristics such as language difficulties, approach problems, help from instructors, and content distribution options. This study discovered these characteristics and their influence on students' commitment to reading. A complete survey is done, concentrating on variables including text difficulty, accessibility problems, teacher encouragement, topic preferences, and study environment. To investigate the data, a multivariable linear regression model was applied to evaluate the connections between independent characteristics and student engagement. Preliminary studies have shown that the combination of these components has a significant impact on the participation decisions. Several advanced machine learning models such as Gradient Boosting Machines (GBM), Random Forest, and Support Vector Machines (SVM) were deployed. The Random Forest and GBM models achieved 95% accuracy in predicting student engagement. Machine Learning models were employed to identify and predict the key cognitive, motivational, and accessibility-related factors. This study emphasized the importance of solving these challenges to improve research skills, improve access, and establish a solid relationship between students and academic resources. In short, we can say that the results give practical advice to educational institutions to design a more effective approach to increase student interaction with research publications.**

Keywords*— Student Engagement, Machine Learning, Educational Barriers, Multivariable Linear Regression.

I. INTRODUCTION

Undergraduate participation in academic research is a significant but inadequately addressed subject in higher education. Students usually experience many problems that impede their capacity to connect well with study information. A multicenter study done in Bangladesh found poor mentoring and restricted access to institutional support systems as key impediments limiting medical students from participating in research [1]. correlates with results from Cureus, which demonstrated that students generally lack expertise in research procedures and confront systemic impediments when trying to perform research independently [5]. Language difficulties and the intricacy of academic works have repeatedly been recognized as important cognitive barriers. Research indicated that several adolescent patients had trouble with academic reading, perceiving both technical jargon and advanced textual organization as relevant concerns [6]. Likewise, engineering students evaluated traditional research material as burdensome owing to their inadequate previous knowledge and the lack of organized aid in interpreting academic articles [2]. These results are part of a wider trend found in STEM occupations, where students have both language and mental health difficulties that make them less confident and motivated. The lack of access to academic materials is another important reason folks don't want to go to college. Students in low-resource contexts sometimes lack access to subscription-based publications or digital libraries. One qualitative study revealed that restricted access to research resources and poor institutional infrastructure were key impediments highlighted by undergraduate dentistry students [7]. Another assessment revealed that institutional gatekeeping and digital inequities disproportionately impact students from poor backgrounds, diminishing their capacity to remain informed of research within their area of study [10]. Motivational and perceptual variables may have a substantial impact on student engagement in research. One study revealed that time constraints, academic pressure, and a lack of co-curricular incentives dramatically lowered student interest in research-related activities among engineering students [8]. Furthermore, students' preconceived conceptions of research as a difficult, inaccessible process prohibited them from interacting with academic themes at all, especially when formal training was missing [9]. While this study indicates a broad diversity of obstacles, most depend on qualitative observations or basic descriptive data. Few have constructed predictive models to measure the degree to which cognitive, motivational, and accessibility-related traits might together predict research involvement. creates a methodological vacuum in the research that hinders the capacity of educators and policymakers to select initiatives. Additionally, only a tiny body of work has studied the use of machine learning (ML) in higher education to tackle such issues. One recent study stated that AI tools like predictive modeling show promise in personalizing training and recognizing troublesome students, but applications to research engagement remain rare [12]. In light of these limitations, the current study contributes to the literature by combining traditional statistical methods (Spearman correlation and multivariable linear regression) with supervised machine learning classifiers (Random Forest, Gradient Boosting Machine, and Support Vector Machine) to predict the likelihood of participating in academic research. The study employs a structured survey incorporating factors anchored on proven barriers—such as text comprehension, technical obstacles, access to resources, and interest in research—to train and analyze prediction models. The purpose is to establish an evidence-based framework that identifies the major drivers of study engagement and explores the consistency of findings across statistics and machine learning methodologies. This dual-method approach provides for both explainability and prediction accuracy—bridging a significant methodological difference in current educational research. Skills of research and increase student engagement in academic research.

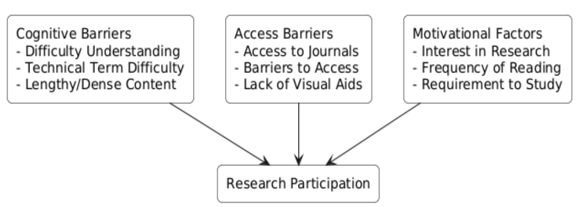

Figure 1: Conceptual Framework

II. Literature Review

Many studies have found that both medical and engineering students face several barriers to participation in academic research. Language difficulty, one of the most frequently cited barriers, was emphasized in a study that found students—particularly in non-native English-speaking regions—struggled with reading and understanding academic research due to the academic language used in the research articles [9]. Complexity of the text and access to resources are also frequently cited as barriers. Notably, one study identified that, along with text complexity, the lack of access to appropriate resources was a key barrier to participation in research by medical students [1]. In addition to language barriers, one of the major challenges is that students lack mentorship and guidance from their instructors. In particular, studies focused on the medical and engineering disciplines indicated that limited support from faculty members had the most impact on student research engagement. For example, one study related to medical students identified that a lack of mentorship and inadequate training in conducting research were two main barriers to students' participation in research activities [5]. In a similar vein, a further study suggested that, while workshops and training run by peers may lower entry barriers for engineering students, a majority described lower levels of formal support and mentorship, that may have limited their participation in research [2]. A related challenge relates to the students' notion of research, particularly when it comes to choosing a suitable topic to empirical investigate. Indeed, when engineering undergraduates were asked about gathering research experience, one study found that many of the students struggled to find and define an appropriate amount of research involvement, indicating that the students' interest and motivation was dampened from engagement [8][12]. These barriers are made even more difficult by students' heavy academic workload and lack of time exploring engagement in research as a possible opportunity. A study identified time constraints as a major barrier to involvement and engagement in academic research for engineering students in particular, who often described challenges that involve competing academic priorities with their research involvement [14]. Another substantial issue relates to access to research papers [6]. Students frequently have difficulty even obtaining the resources they would need to include in their process of obtaining incremental knowledge, whether that is limited academic access to databases and online journals, or limitations to expensive journal subscriptions. One study examined each of these challenges and found that having limited access to scholarly resources was very detrimental in students' participation in academic research [12]. The deprivation of access is especially true in resource-poor contexts, where students struggle to obtain the materials needed to find out what is current in the research for their field of study [10]. In addition, another publication noted that students in general, and women students in particular, face barriers to attaining research experiences that are due to societal expectations and a lack of institutional support [13]. Also, teacher support is crucial to generating research engagement from students. In one study, medical students indicated they were more likely to engage in research activities when their teachers encouraged and gave them direction [1]. Many students suggest that teacher support is often lacking, especially at institutions where research is limited. A lack of support from faculty has neglected to build confidence in engaging with the subject matter, along with students' ability to believe they could be part of producing academic knowledge [11].

Building upon these well-documented issues, research has applied machine learning and artificial intelligence technologies to better analyze and anticipate student involvement habits. Behavioral data is used from Moodle and applied interpretable ML models to demonstrate that session length, frequency, and variety were substantially linked with student involvement patterns [15]. Similarly, another technique offered a sentiment-based engagement measure leveraging large language models (LLMs) to evaluate textual input in e-learning contexts, allowing scalable, real-time evaluations of engagement [16]. The relevance of early identification was underlined in a study that employed ensemble ML approaches to identify at-risk children based on demographic and behavioral characteristics, indicating key moments for academic intervention [17]. A more specialized research employed contrastive learning to build ordinal involvement levels from student facial expressions and body language, demonstrating a complex way to capture engagement beyond binary categories [18].

Even with this issue of non-reading, the increasing need for reading for academic research and educational purposes is spurring many academic institutions to find meaningful solutions. The academic work presented in this paper extends the current literature on factors such as text difficulty, accessibility challenges, teacher encouragement, and topic preferences using advanced machine learning approaches. We will employ models, such as Gradient Boosting Machines, Random Forest, and Support Vector Machines, to identify important factors influencing student engagement with research papers. We aim to provide action plans that educational institutions can implement in support of student engagement with academic research and creating meaningful approaches to teaching research skills.

III. Methodology

A. Data Collection

A systematic survey has been conducted with undergraduate university students to examine their attitudes towards participation in research-related activities and the challenges associated with them. The questionnaire asked pertaining to 12 variables associated with understanding, motivation, and access. The responses were measured on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree) and consists of independent variables pertained to:

Cognitive factors: Difficulty Understanding, Difficulty with Technical Terms, Lengthy and Dense Text, STEM Research Methods Understanding, and Overall Understanding.

Environmental/accessibility factors: Lack of Visual Presentation, Difficulty of Access, Barriers to Access.

Motivational factors: Requirement to Study, How Often Read, Participation in Research, and Interest in Research.

The dataset included 400 responses of the university students age from 18 to 24 years. The sample size of 400 was selected based on Cochran's formula with a 95% confidence level and a 5% margin of error, providing appropriate power to detect statistically significant effects in regression and classification models. All categorical variables were encoded numerically. The dependent variable, Research Participation, was binary (0 = No, 1 = Yes) which is the target variable for the study.

B. Statistical Analysis

A two-step statistical approach was applied:

1. Spearman Correlation Analysis was conducted to measure monotonic relationships between independent variables and the target variable. Spearman's ρ is calculated as:

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

Where d_i is the difference between ranks of each pair of observations, and n is the number of observations.

2. Multivariable Linear Regression was employed to determine the collective influence of independent variables on *Research Participation*. The regression model is expressed as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \varepsilon$$

Where:

- Y is the predicted value of research participation,
- β_0 is the interception,
- β_i are the coefficients for each predictor,
- ε is the error term.

C. Machine Learning Classification

To further validate the predictive capacity of key variables, the dataset was used to train three classification models:

1. Random Forest (RF)

An ensemble method that constructs multiple decision trees and outputs the mode of their predictions.

$$\hat{y} = mode(h_1(x), h_2(x), \ldots, h_T(x),$$

2. Gradient Boosting Machine (GBM)

A boosting method that sequentially adds weak learners to minimize prediction error.

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$$

where F_m is the boosted model at iteration m, h_m, is the new weak learner, and γ_m is the learning rate.

3. Support Vector Machine (SVM)

A supervised model that finds the hyperplane maximizing the margin between classes:

$$min \frac{1}{2}\|w\|^2 \; subject \; to \; y_i(w^T x_i + b) \geq 1$$

where w is the weight vector and b is the bias.

D. Evaluation Metrics

Each model was trained using an 80/20 train-test split and evaluated using:

- Accuracy: $\frac{TP+TN}{TP+TN+FP+FN}$
- Precision: $\frac{TP}{TP+FP}$
- Recall: $\frac{TP}{TP+FN}$
- F1 Score: Harmonic mean of Precision and Recall.

IV. RESULT

In order to capture the comprehensive picture of the factors that influence undergraduate students' engagement with academic research, both statistical and machine learning methods were utilized on the data collected from the survey. The findings are presented in two sections: Statistical Analysis and Machine Learning Analysis.

A. Statistical Analysis

1) Correlation Analysis

A Spearman correlation analysis was conducted to assess the relationships among cognitive, structural, motivational, and accessibility-related variables with research participation. Table I summarizes key correlation coefficients.

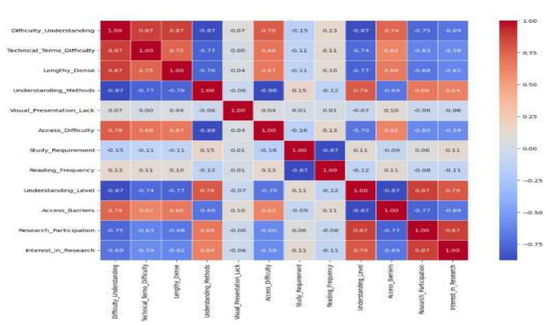

Fig 2: Correlation heatmap

TABLE I. SPEARMAN CORRELATION WITH RESEARCH PARTICIPATION

Variable	Correlation Coefficient (ρ)
Understanding Level	0.874
Interest in Research	0.788
Understanding Research Methods	0.676
Difficulty Understanding	-0.754
Lengthy or Dense Content	-0.681
Technical Terms Difficulty	-0.628
Access Difficulty	-0.587

The results suggest that comprehension variables like Understanding Level and Understanding of Research Methods are strongly positively correlated to participating in research, which indicates that clarity of cognition is an important element to participating in research. On the contrary, Difficulty of Understanding, Difficulty with Technical Words, and Amount of Content are significantly negatively related to participating in research, which suggests relevant language and structural challenges.

2) Regression Analysis

A multivariable linear regression model was created with Research Participation (binary: 1 = Yes, 0 = No) as the outcome variable. The independent variables were coded from the Likert-scale survey items. The model produced an $R^2 = 0.789$ indicating approximately 79% of the variance in research participation is explained by the included variables.

These findings align with the correlation findings, supporting the conclusion that cognitive and access barriers are most significant to participation. The other factors, such as visual presentation or study demands, had small coefficients and were not included in the prediction model, primarily because they were not statistically significant.

TABLE II. REGRESSION COEFFICIENTS (STANDARDIZED)

Predictor	Coefficient (β)
Understanding Level	0.321
Understanding Research Methods	0.283
Difficulty Understanding	-0.296
Access Difficulty	-0.217
Technical Terms Difficulty	-0.198

The following variables can be formulated into a regression equation as follows:

Predicted Research Participation = .321 × Understanding Level + .283 × Understanding Methods −

$.296 \times \text{Difficulties with Understanding} - .217 \times \text{Difficulties with Access} - .198 \times \text{Technical Terms} + \varepsilon$

Where ε represents the error term. In conclusion, the effects point to the improvements cognitive supports and simplifying research content are likely to significantly increase participation rates for students.

B. Machine Learning Analysis

In order to confirm and augment the statistical findings, three supervised classification models were used: random forest (RF), gradient boosting machine (GBM), and support vector machine (SVM). The dataset was separated into 80% training and 20% testing. The target variable was binary research participation. The metrics used for estimation were accuracy, precision, recall, and F1-score.

TABLE III. MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.950	0.944	0.944	0.944
Gradient Boosting	0.950	0.944	0.944	0.944
Support Vector Machine	0.938	0.970	0.889	0.928

The Random Forest and Gradient Boosting models both performed equally well at a high accuracy of 95%, suggesting that both models were a good classifier. The SVM was slightly lower accuracy at 93.8% but performed better, in terms of precision (96.97%) than either model indicating that it performed better at avoiding false positives.

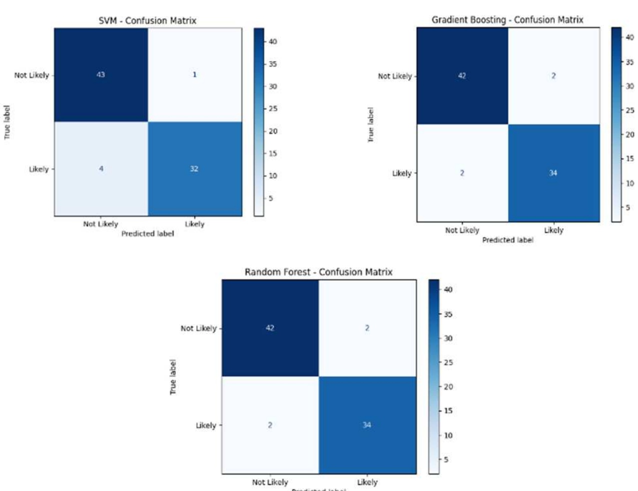

Fig 3: Confusion Matrix

The machine learning models displayed extremely high accuracy, presumably due to the homogeneous sample and well-designed, targeted survey settings. Confusion matrices exhibited balanced predictions, with minimal false positives and negatives across all models. These were checked before supplying aggregate metrics to confirm classification quality. Synthesis and Implications

C. Synthesis and Implications

Both statistical and machine learning analyses converge on support for the overarching hypothesis: cognitive clarity, methodological understanding, and access are instrumental in supporting student engagement in research publications. Within the regression model, the high R^2 value and all machine learning analyses indicated a strong classification accuracy for student engagement. Together, these findings exemplify that student engagement can be predicted and potentially increased through interventions. It is recommended that educational stakeholders address and/or prioritize the following:

- Provide research methodology courses at the beginning stage of curricula,
- Produce plain-language versions of research materials,
- Improve access to journals and academic research content digitally.

By addressing at least a few modifiable boundaries identified in some of the research study passages, colleges and universities may be able to advance a stronger research culture in undergraduates. The congruence between statistical regression and ML classifiers enhances the resilience of essential variables like knowledge level and access constraints. This consistency underscores the dependability of utilizing both methodologies to detect and foresee constraints to undergraduate research engagement.

V. CONCLUSION

This study presents an in-depth examination of the various elements that influence undergraduate students' participation in academic research. Investigating students' research engagement, we utilized both statistical and machine learning approaches to identify impediments associated with students' research engagement. Results indicated that cognitive clarity, understanding of the research process, and accessibility were key influences for students' engagement with research. More specifically, the correlation analysis and regression model pointed to several clear relationships among comprehension-related variables such as understanding and research methods, as well as accessibility issues such as access difficulty and use of technical terminology. Finally, the machine learning models, specifically Random Forest and Gradient Boosting, provided good classification accuracy, lending further support to the importance of the findings.

The results highlight the value of universities emphasizing cognitive support systems for undergraduates, choices looking to reduce the burden on undergraduate students engagement of research, task access to academic resources. By addressing these barriers, universities will be able to nurture a more robust research culture among undergraduates, along with improvements in academic accomplishment, engagement, and contributions to research. The current study also offers important considerations for curriculum designers, faculty, and stakeholders to support increased engagement of students towards academic research.

VI. Limitation

Although this study has shed light on the obstacles to student engagement with academic research, it is important to recognize a number of limitations. The first of which is that it is based on survey data, which is grounded on self-reported measures. Inherent in self-reported data is the possibility of bias in respondents' answers. The sample itself also might not be fully representative of the breadth of students across disciplines and institutions, which impacts the extent to which the findings can be generalized. Future studies would benefit from a more diverse sample and longitudinal data to examine the sustained impact of tackling the barriers identified here. A final limitation is that the machine learning model produced robust predictions regarding student engagement with academic research but does not encompass the nuances involved in human behavior and decision making, in which case further development and validation of the models is needed.

VII. Acknowledgement

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